

LIN ZHANG

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SUMMARY

I hold a Ph.D. from the Yamagishi Lab at the National Institute of Informatics, Japan. Prior to that, I completed my master's degree at Tianjin University, China. I have served as a research assistant or visiting researcher at three academic institutions across China, Japan, and Europe. My research focuses on **speech security and privacy**, specifically on **anti-spoofing (creator of [Partial Spoof](#))**, speech anonymization, and speaker recognition. I have collaborated with researchers from around the world and published more than 30 papers in top-tier speech-related conferences and journals, including INTERSPEECH, ICASSP, TASLP, and TIFS.

EDUCATION

Ph.D The Graduate University for Advanced Studies (SOKENDAI) <i>Supervisor:</i> Prof. Junichi Yamagishi; <i>Mentor:</i> Dr. Xin Wang, Dr. Erica Cooper <i>Focus:</i> Anti-spoofing (Partial Spoof), Speaker Recognition <i>Thesis:</i> "Whether, When, What": Detection, Localization, and Diarization of Partially Spoofed Audio	Oct. 2020 – Mar. 2024 Tokyo, Japan
M.Sc Tianjin University <i>Supervisor:</i> Prof. Kiyoshi Honda and Prof. Jianguo Wei <i>Focus:</i> Speaker Recognition, Speech Production, Speech Recognition <i>Thesis:</i> Study on Static Individual Characteristics for Speaker Recognition	Sep. 2017 – Jan. 2020 Tianjin, China
B.Sc North China University of Water Resources and Electric Power <i>Major:</i> Software Engineering, 1/102 <i>Thesis:</i> Research on Automatic Speech Recognition Based on Deep Learning	Sep. 2013 – Jul. 2017 Henan, China

WORK EXPERIENCE

Postdoctoral Fellow Johns Hopkins University (Dr. Nicholas Andrews, Dr. Matthew Wiesner, Dr. Leibny Paola Garcia, Prof. Sanjeev Khudanpur) Speech Security and Privacy	April 2025 – present Baltimore, United States
Research Worker Brno University of Technology (Prof. Lukáš Burget) Anti-spoofing, Speaker Verification, Speaker Diarization.	July 2024 – March 2025 Brno, Czech Republic
Research Assistant National Institute of Informatics (Prof. Junichi Yamagishi) - Defined a new spoofing scenario – 'Partial Spoof,' - Built a database PartialSpoof with more than <u>5k+</u> downloading, - Defined tasks and measurements for Partial Spoof, - Proposed several benchmark countermeasures.	Oct. 2020 – Mar. 2024 Tokyo, Japan
Visitor Brno University of Technology (Prof. Lukáš Burget, Dr. Mireia Diez) Best Paper of Odyssey 2024 - Applied the variational information bottleneck to investigate the essential information required for EEND-EDA.	May 2023 – Oct. 2023 Brno, Czech Republic
Research Assistant Duke Kunshan University (Prof. Ming Li) Data selection, knowledge-based feature extraction, model structure for an autism project.	Feb. 2020 – Aug. 2020 Jiangsu, China

SKILLS

Languages: Mandarin (Native), English

Programming: Python, MATLAB, Shell, C/C++, Java, JavaScript, HTML5, SQL

Toolkits: Pytorch, Kaldi, Scikit-learn

PUBLICATION

Journal

10. Jiawei Du, Xuanjun Chen, Haibin Wu, **Lin Zhang**, I-Ming Lin, I-Hsiang Chiu, Wenze Ren, Yuan Tseng, Yu Tsao, Jyh-Shing Roger Jang, Hung-yi Lee. (2025) "CodecFake+: A Large-Scale Neural Audio Codec-Based Deepfake Speech Dataset" (submitted to IEEE Transactions on Audio, Speech and Language Processing)
9. Ruiteng Zhang, Jianguo Wei, Xugang Lu, **Lin Zhang**, Di Jin, Wenhuan Lu, Junhai Xu. (2025) "Multi-Sinkhorn Teacher Knowledge Aggregation Framework for Adaptive Audio Anti-Spoofing" in IEEE Transactions on Audio, Speech and Language Processing, vol. 33, pp. 3850-3865.
8. Ruiteng Zhang, Jianguo Wei, Xugang Lu, **Lin Zhang**, Di Jin, Junhai Xu, Wenhuan Lu. (2025) "SHDA: Sinkhorn Domain Attention for Cross-Domain Audio Anti-Spoofing" IEEE Transactions on Information Forensics and Security, vol. 20, pp. 6474-6489, 2025, doi: 10.1109/TIFS.2025.3576576
7. Ruiteng Zhang, Jianguo Wei, Xugang Lu, Yongwei Li, Wenhuan Lu, **Lin Zhang**, and Junhai Xu. (2025) An Adaptation Framework with Unified Embedding Reconstruction for Cross-Corpus Speech Emotion Recognition, Applied Soft Computing Journal, vol. 174, pp. 112948.
6. Xinlei Ma, Ruiteng Zhang, Jianguo Wei, Xugang Lu, Junhai Xu, **Lin Zhang**, and Wenhuan Lu. (2025) "Self-distillation-based domain exploration for source speaker verification under spoofed speech from unknown voice conversion." Speech Communication, vol 167: 103153.
5. Ruiteng Zhang, Jianguo Wei, Xugang Lu, Junhai Xu, Yongwei Li, Di Jin, **Lin Zhang**, Wenhuan Lu (2025). "Optimal Transport with Class Structure Exploration for Cross-Domain Speech Emotion Recognition." in IEEE/ACM Transactions on Audio, Speech and Language Processing, vol. 33, pp. 37-53, 2025
4. Ruiteng Zhang, Jianguo Wei, Xugang Lu, Wenhuan Lu, Di Jin, **Lin Zhang**, and Junhai Xu. (2024). "Unsupervised Adaptive Speaker Recognition by Coupling-Regularized Optimal Transport." IEEE/ACM Transactions on Audio, Speech, and Language Processing. vol. 32, pp. 3603-3617.
3. Ruiteng Zhang, Jianguo Wei, Xugang Lu, Wenhuan Lu, Di Jin, **Lin Zhang**, Yantao Ji, and Junhai Xu. (2023). "Self-supervised learning based domain regularization for mask-wearing speaker verification." Speech Communication: 102953.
2. Ruiteng Zhang, Jianguo Wei, Xugang Lu, Wenhuan Lu, Di Jin, **Lin Zhang**, Junhai Xu, and Jianwu Dang. "TMS: Temporal multi-scale in time-delay neural network for speaker verification." Applied Intelligence 53, no. 22 (2023): 26497-26517.
1. **Lin Zhang**, Xin Wang, Erica Cooper, Nicholas Evans, Junichi Yamagishi. (2023). The PartialSpoof Database and Countermeasures for the Detection of Short Fake Speech Segments Embedded in an Utterance, IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 31, pp. 813-825.

Conference

24. **Lin Zhang**, Xin Wang, Johan Rohdin, Junyi Peng, Tianchi Liu, You Zhang, Hieu-Thi Luong, Shuai Wang, Anna Silnova, Chengdong Liang, Nicholas Evans. "WeDefense: A Toolkit to Defend Against Fake Audio" (in preparation)
23. I-Ming Lin, Xuanjun Chen, **Lin Zhang**, Haibin Wu, Hung-yi Lee, Jyh-Shing Roger Jang. "Towards Generalized Source Tracing for Codec-Based Deepfake Speech" (accepted to Proc. IEEE ASRU 2025)
22. Ashi Garg, Zexin Cai, **Lin Zhang**, Henry Li Xinyuan, Leibny Paola Garcia Perera, Kevin Duh, Sanjeev Khudanpur, Matthew Wiesner, Nicholas Andrews "ShiftySpeech: A Large-Scale Synthetic Speech Dataset with Distribution Shifts" (Submitted to NeurIPS 2025 Datasets and Benchmarks Track)
21. You Zhang, Baotong Tian, **Lin Zhang**, Zhiyao Duan. "PartialEdit: Identifying Partial Deepfakes in the Era of Neural Speech Editing," in Proc. Interspeech 2025, 5353-5357.

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20. Xuanjun Chen, I-Ming Lin, **Lin Zhang**, Jiawei Du, Haibin Wu, Hung-yi Lee, Jyh-Shing Roger Jang. "Codec-Based Deepfake Source Tracing via Neural Audio Codec Taxonomy," in Proc. Interspeech 2025, 1538-1542.
19. Sara Barahona, Anna Silnova, Ladislav Mošner, Junyi Peng, Oldřich Plchot, Johan Rohdin, **Lin Zhang**, et. al. "Analysis of ABC Frontend Audio Systems for the NIST-SRE24" in Proc. Interspeech 2025, 5763-5767.
18. Hieu-Thi Luong, Haoyang Li, **Lin Zhang**, Kong Aik Lee, Eng Siong Chng. "LlamaPartialSpoof: An LLM-Driven Fake Speech Dataset Simulating Disinformation Generation," in Proc. ICASSP 2025, pp. 1-5.
17. Junyi Peng, Ladislav Mošner, **Lin Zhang**, Oldřich Plchot, Themis Stafylakis, et al. "CA-MHFA: A Context-Aware Multi-Head Factorized Attentive Pooling for SSL-Based Speaker Verification," in Proc. ICASSP 2025, pp. 1-5.
16. Xiaohuan Chen, Wenhuan Lu, Ruiteng Zhang, Junhai Xu, Xugang Lu, **Lin Zhang**, Jianguo Wei. "Continual Unsupervised Domain Adaptation for Audio Deepfake Detection," in Proc. ICASSP 2025, pp. 1-5.
15. Tianchi Liu, **Lin Zhang**, Rohan Kumar Das, Yi Ma, Ruijie Tao, Haizhou Li. "How Do Neural Spoofing Countermeasures Detect Partially Spoofed Audio?" in Proc. Interspeech 2024, pp. 1105-1109.
14. Johan Rohdin, **Lin Zhang**, Oldřich Plchot, et al. "BUT Systems and Analyses for the ASVspoof 5 Challenge," in Proc. ASVspoof 2024 Workshop, pp. 24-31.
13. **Lin Zhang**, Themis Stafylakis, Lukáš Burget, Federico Landini, Mireia Diez, Anna Silnova. "Do End-to-End Neural Diarization Attractors Need to Encode Speaker Characteristic Information?" in Proc. Odyssey 2024, pp. 123-130.
12. **Lin Zhang**, Xin Wang, Erica Cooper, Mireia Diez, Federico Landini, Nicholas Evans, Junichi Yamagishi. "Spoof Diarization: "What Spoofed When" in Partially Spoofed Audio," in Proc. Interspeech 2024, pp. 502-506.
11. Ruiteng Zhang, Jianguo Wei, Xugang Lu, Wenhuan Lu, Di Jin, **Lin Zhang**, Junhai Xu. "Optimal Transport with a Diversified Memory Bank for Cross-Domain Speaker Verification," in Proc. ICASSP 2023, pp. 1-5.
10. **Lin Zhang**, Xin Wang, Erica Cooper, Nicholas Evans, Junichi Yamagishi. "Range-Based Equal Error Rate for Spoof Localization," in Proc. Interspeech 2023, pp. 3212-3216.
9. Xiaohui Liu, Meng Liu, **Lin Zhang**, et al., "Deep Spectro-temporal Artifacts for Detecting Synthesized Speech," in Proc. DDAM 2022, pp. 69-75.
8. Linjuan Zhang, Kong Aik Lee, **Lin Zhang**, Longbiao Wang, Baoning Niu. "CpAug: Refining Copy-Paste Augmentation for Speech Anti-Spoofing," in Proc. ICASSP 2022, pp. 7082-7086.
7. Chang Zeng, **Lin Zhang**, Meng Liu, Junichi Yamagishi. "Spoofing-Aware Attention-based ASV Back-end with Multiple Enrollment Utterances and a Sampling Strategy for the SASV Challenge 2022," in Proc. Interspeech 2022, pp. 2883-2887.
6. Kai Li, Li, S., Lu, X., Akagi, M., Liu, M., **Lin Zhang**, Zeng, C., Wang, L., Dang, J., Unoki, M. "Data Augmentation Using McAdams-Coefficient-Based Speaker Anonymization for Fake Audio Detection," in Proc. Interspeech 2022, pp. 664-668.
5. Ruiteng Zhang, Jianguo Wei, Wenhuan Lu, **Lin Zhang**, Yantao Ji, Junhai Xu, Xugang Lu. "CS-REP: Making Speaker Verification Networks Embrace Re-Parameterization," in Proc. ICASSP 2022, pp. 7082-7086.
4. **Lin Zhang**, Xin Wang, Erica Cooper, Junichi Yamagishi, et al. "An Initial Investigation for Detecting Partially Spoofed Audio," in Proc. Interspeech 2021, pp. 4264-4268.
3. **Lin Zhang**, Xin Wang, Erica Cooper, Junichi Yamagishi. "Multi-task Learning in Utterance-level and Segmental-level Spoof Detection," in Proc. ASVspoof Workshop 2021, pp. 9-15.
2. **Lin Zhang**, Kiyoshi Honda, Jianguo Wei, Seiji Adachi. "Regional Resonance of the Lower Vocal Tract and its Contribution to Speaker Characteristics," in Proc. Interspeech 2020, pp. 1391-1395.
1. Ruiteng Zhang, Jianguo Wei, Wenhuan Lu, Longbiao Wang, Meng Liu, **Lin Zhang**, Jiayu Jin, Junhai Xu. "ARET: Aggregated Residual Extended Time-Delay Neural Networks for Speaker Verification," in Proc. Interspeech 2020, pp. 946-950.

(Nov. 2025)

Patent

2. Speaker Recognition Based on High and Low Frequency Dynamic and Static Features, by Jianguo Wei, Lin Zhang, Ruiteng Zhang. (September 15, 2023) [CN110931022A](#)
1. Cloud language ability evaluation system and wearable recording terminal, by Ming Li, Lin Zhang, Qingjian Lin. (April 30, 2024) [CN112750465A](#)

INVITED TALKS

Academia: IEEE SPS (online, Nov. 2025), UEF (Finland, Mar. 2025), EURECOM (France, Feb. 2025), ISCA-SPSC (online, July 2024), BUT (Czechia, June, 2023)
Industry: Reality Defender (online, Aug. 2025), Huawei Shield Trustworthy Lab (online, May 2023)

REVIEWER

IEEE Transactions on Information Forensics and Security (IF: 7.178)	2023 – Present
IEEE/ACM transactions on audio, speech, and language processing (IF: 4.364)	2023 – Present
ACM Transactions on Intelligent Systems and Technology (IF: 4.031)	2023 – Present
Computer Speech & Language (IF: 3.1)	2024 – Present
Interspeech, ICASSP, IEEE SLT, IEEE ASRU	2024 – Present
VoicePrivacy, Symposium on Security and Privacy in Speech Communication	2024 – Present

MENTORING

Sandra Arcos-Holzinger, University of Melbourne, Australia & Johns hopkins University, USA	2025 – Present
Kai Tan, Johns hopkins University, USA	2025 – Present
Ruiteng Zhang, Tianjin University, China	2019 – 2020

SERVICE

Organizer: IEEE ASRU 2025 special session about [Frontiers in Deepfake Voice Detection and Beyond](#)
Joint Workshop of VoicePersonae and ASVspoof 2023
Member: [SPSC-SIG Appointed Team 2025- Present](#)
Volunteer: INTERSPEECH 2023-2024, HHME 2018, ZHENG-KAI International Marathon 2015

HONORS AND AWARDS

AWM Travel Grants, Association for Women in Mathematics	2025
Best Paper Award, Odyssey	2024
Special Researcher Scholarship, SOKENDAI	2021 - 2024
Superior Graduate of Henan Province, The Education Department of Henan Province	2017
National Scholarship, Chinese Ministry of Education	2014, 2016